

Transactions & ACID Properties

Lab 1:

Ex1:

- Step 1: Check initial status

```
1 SELECT rental_id, return_date
2 FROM rental
3 WHERE rental_id = 2;
```

Data Output Messages Notifications

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rental_id [PK] integer	return_date timestamp without time zone
2	2005-05-28 19:40:33

- Step 2: Execute a transaction block designed to trigger an error.

```
5 BEGIN;
6
7 -- 1. Cập nhật ngày trả đĩa
8 UPDATE rental
9 SET return_date = NOW()
10 WHERE rental_id = 2;
11
12 -- 2. Cố tình gây lỗi sai định dạng ngày ('WRONG_DATE_FORMAT')
13 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
14 VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
15
16 COMMIT;
```

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WARNING: there is already a transaction in progress

ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
LINE 10: VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
^

SQL state: 22007
Character: 265

- Step 3: Rerun the check command

19	SELECT return_date
20	FROM rental
21	WHERE rental_id = 2;

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	return_date timestamp without time zone
1	2005-05-28 19:40:33

1. Is the **return_date** updated? → No.
2. Why not? → When the **INSERT** statement encounters a data type error, PostgreSQL immediately puts the transaction into an **aborted state**. At this point, the **COMMIT** statement at the end effectively becomes a **ROLLBACK**, undoing the previous **UPDATE** statement and preserving the original data according to the principle of **Atomicity (All or Nothing)**.

Ex 2:

22	BEGIN;
23	
24	-- Bước 1: Tạo nhân viên mới (chưa gắn cửa hàng cụ thể hoặc gắn tạm)
25	INSERT INTO staff (first_name, last_name, address_id, email, store_id, active, username)
26	VALUES ('Alex', 'Brown', 1, 'alex.brown@sakilastaff.com', 1, true, 'alex_b');
27	
28	-- Bước 2: Tạo điểm đánh dấu an toàn (SAVEPOINT)
29	SAVEPOINT staff_created;
30	
31	-- Bước 3: Thử gắn nhân viên vào store_id = 9999 không tồn tại (sẽ gây lỗi vi phạm Foreign Key)
32	UPDATE staff
33	SET store_id = 9999
34	WHERE username = 'alex_b';
35	
36	-- Bước 4: Khi bước 3 lỗi, quay lui về mốc an toàn và gắn vào Store 1 hợp lệ
37	ROLLBACK TO staff_created;
38	
39	UPDATE staff
40	SET store_id = 1
41	WHERE username = 'alex_b';
42	
43	-- Bước 5: Xác nhận lưu toàn bộ thay đổi hợp lệ
44	COMMIT;
45	

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COMMIT
Query returned successfully in 268 msec.

- Run the query to look up the new staff:

```
47
48 SELECT staff_id, first_name, last_name, store_id, username
49 FROM staff
50 WHERE username = 'alex_b';
```

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	staff_id [PK] integer 	first_name character varying (45) 	last_name character varying (45) 	store_id smallint 	username character varying (16) 
1	3	Alex	Brown	1	alex_b

The staff Alex Brown was successfully created in the system with store_id = 1, despite the fact that the previous operation to assign them to Store 9999 had resulted in an error.